

**FORM TP 2005198**



TEST CODE **02120020**

MAY/JUNE 2005

**CARIBBEAN EXAMINATIONS COUNCIL  
ADVANCED PROFICIENCY EXAMINATION**

**ENVIRONMENTAL SCIENCE**

**UNIT 1: ECOLOGY, PEOPLE AND NATURAL RESOURCE USE**

**PAPER 02**

*2 hours 30 minutes*

**27 MAY 2005 (p.m.)**

**READ THE FOLLOWING INSTRUCTIONS CAREFULLY**

1. **DO NOT** open this examination paper until instructed to do so.
2. This paper consists of **NINE** questions, **THREE** from each Module.
3. Candidates **MUST** answer **SIX** questions, **TWO** from **EACH** Module.
4. Write your answers in the answer booklet provided.

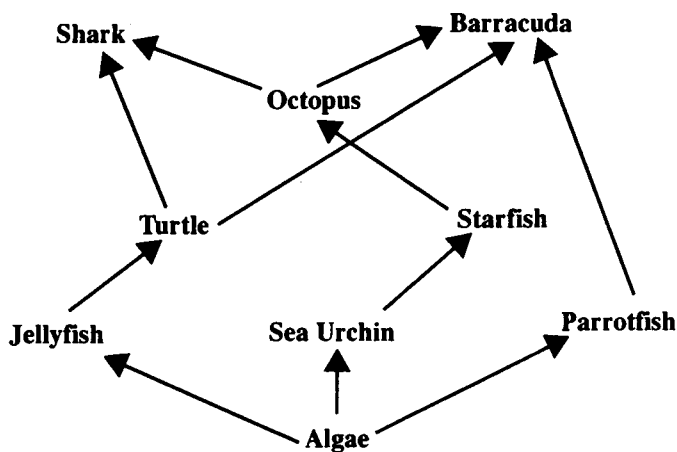
## MODULE 1

Answer ANY TWO questions.

1. (a) In the carbon cycle, there are carbon sinks and carbon sources. Using an example of EACH, distinguish between 'carbon sinks' and 'carbon sources'. [ 4 marks]
- (b) With the use of an annotated diagram, outline
- (i) the movement of carbon between the atmosphere and the biotic components of the terrestrial ecosystem [ 5 marks]
  - (ii) the phosphorous cycle. [ 6 marks]
- (c) (i) Identify the MAJOR difference between the carbon cycle and the phosphorous cycle. [ 1 mark ]
- (ii) Outline the MAJOR consequence of the difference between both cycles indicated in (c) (i). [ 4 marks]

**Total 20 marks**

2. Figure 1 below shows a food web for a marine ecosystem.



**Figure 1. A food web for a marine ecosystem**

- (a) Group the organisms shown in the food web in their respective trophic levels. [ 5 marks]
- (b) Explain how energy becomes incorporated in this ecosystem. [ 5 marks]
- (c) Only a small percentage of energy incorporated into the ecosystem is available to the barracuda. Suggest TWO reasons for this. [ 4 marks]
- (d) Explain how this community might be altered if the population of sea urchins died out. [ 6 marks]

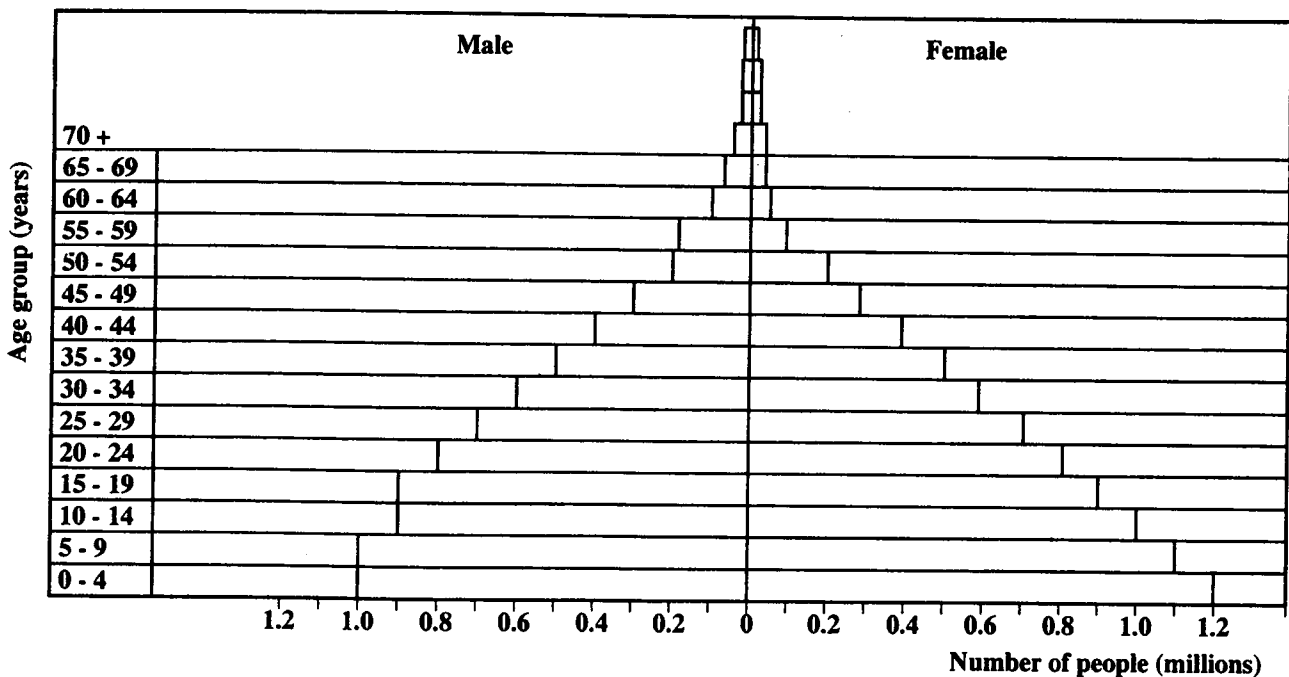
**Total 20 marks**

3. (a) Using specific examples, describe TWO impacts of humans on ecosystems. [ 8 marks]
- (b) Outline FOUR reasons why it is important for human beings to maintain the integrity of ecosystems. [12 marks]
- Total 20 marks**

## MODULE 2

Answer ANY TWO questions.

4. The age structure diagram in Figure 2 is that of Country X with a total population of sixteen million.



**Figure 2. Age structure diagram for Country X**

- (a) Calculate the percentage of the population in the age group ranging from 0 to 19 years. [ 5 marks]
- (b) Describe how the growth rate of the population will be determined by the age group ranging from 0 to 19 years. [ 6 marks]
- (c) Outline the impact on the environment that this population could have if it grows at the rate discussed in (b) above. [ 7 marks]
- (d) State TWO characteristics of the population that can be inferred from the age structure diagram. [ 2 marks]

**Total 20 marks**

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5. Table 1 shows some selected demographic statistics for Ethiopia.

**TABLE 1: SELECTED DEMOGRAPHIC STATISTICS FOR ETHIOPIA**

| Country  | Total Fertility Rate | Life Expectancy at Birth | Women Using Contraceptives (%) |
|----------|----------------------|--------------------------|--------------------------------|
| Ethiopia | 7.0                  | 42                       | 3                              |

- (a) Define the terms
- (i) total fertility rate [ 2 marks]
- (ii) life expectancy. [ 2 marks]
- (b) With reference to Table 1, outline the relationship between the total fertility rate and the life expectancy for the Ethiopian population. [ 6 marks]
- (c) Suggest TWO reasons for the low percentage of Ethiopian women using contraceptives. [ 6 marks]
- (d) Suggest an approach aimed at decreasing the fertility rate of Ethiopian women. [ 4 marks]

**Total 20 marks**

6. (a) List SIX general characteristics of persons living in poverty. [ 3 marks]
- (b) State and define TWO international indices used to rank the development status of countries. [ 4 marks]
- (c) Population growth in a country will lead to an increase in poverty levels. Do you agree with this statement? Justify your answer. [ 6 marks]
- (d) Discuss how increasing levels of poverty can hinder the ability of a country to achieve sustainable development. [ 7 marks]

**Total 20 marks**

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### MODULE 3

Answer ANY TWO questions.

7. In many Caribbean countries, fishing is an important economic and subsistence activity. There are claims that many of the fishing grounds and habitats have been damaged or polluted and there are fears that the fishing industry will collapse in the near future. The governments, fishermen and fishing industry officials have been discussing issues related to the industry and are seeking to implement appropriate actions to prevent its collapse.

The graphs in Figures 3 and 4 show the variation in the quantity of fish caught and the number of boats engaged in fishing over the last ten years.

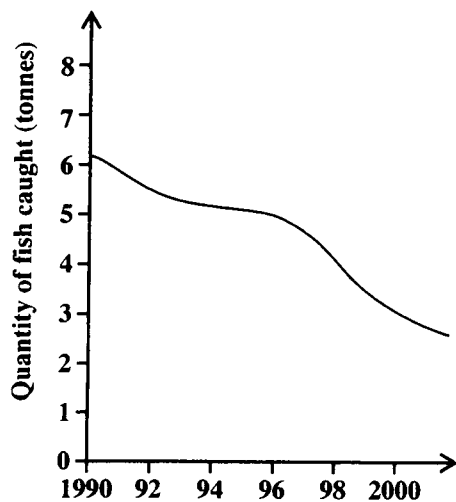


Figure 3. Quantity of fish caught

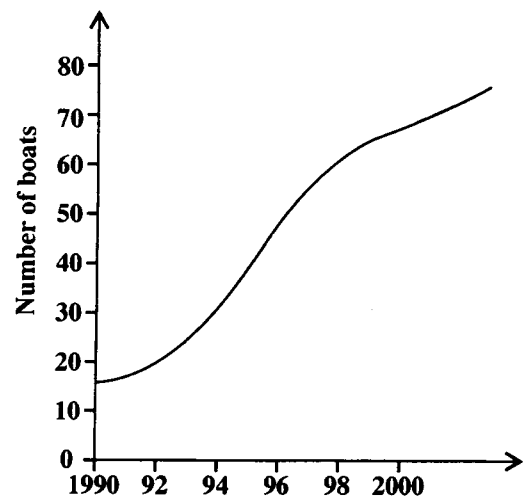


Figure 4. Number of boats engaged

- (a) Describe the trends indicated in Figures 3 and 4. [ 3 marks]
- (b) Could the problems experienced by the fishing industry have been prevented? Justify your response. [ 5 marks]
- (c) Outline THREE approaches to deal with the current problems being experienced in the fishing industry. [ 6 marks]
- (d) Assess the effectiveness of TWO of the approaches outlined in (c) above in a Caribbean context. [ 6 marks]

**Total 20 marks**

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8. (a) Describe TWO functions of mangroves in Caribbean ecosystems. [ 8 marks]
- (b) Discuss TWO main impacts on mangroves in Caribbean ecosystems which result from developmental activities. [ 8 marks]
- (c) Outline TWO appropriate methods of conserving mangroves in a Caribbean country. [ 4 marks]

**Total 20 marks**

9. A major deposit of gold has been discovered in a largely forested and hilly location in a Caribbean country. The Government of this country is encouraging local and foreign investment to establish gold mining operations for extracting the gold. Adjacent to the deposit is a river which supplies a nearby community with water for domestic and agricultural needs.

Discuss FOUR factors that should be considered in determining the feasibility of mining this resource. **Total 20 marks**

**END OF TEST**